

Free-Field Computation of the $\mathbf{Vir} \oplus \mathbf{Vir}$ Branching Coefficient

Highest-vector prescription, norm, and three-point numerator

Machine working note

August 16, 2026

Abstract

The human note reduces the all-NS superconformal block to two ordinary Virasoro blocks and leaves one quantity undetermined: the normalized three-point coefficient of the $\mathbf{Vir} \oplus \mathbf{Vir}$ highest-weight vectors v_n . Here we compute that quantity from the free-field prescription of arXiv:1111.2803. Writing $k = 2n \in \mathbb{Z}$, the vector is a finite Fermi-sea product of the modes $\chi_r = f_r - i\partial_r$. Its three-point numerator is the factorized blow-up factor l of that paper. Specializing the same formula to the identity operator gives the Shapovalov norm. Combining the two produces a closed product formula for the squared branching coefficient that appears in the human note. No Whittaker vector is used.

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1 NS free-field realization

We first fix the realization in which the branching vector will be computed. Set

$$Q = b + b^{-1}, \quad c = \frac{3}{2} + 3Q^2, \quad h(P) = \frac{1}{2} \left(\frac{Q^2}{4} - P^2 \right). \quad (1)$$

All modes in this section are NS modes: $m \in \mathbb{Z}$ and $r \in \mathbb{Z} + \frac{1}{2}$.

1.1 Oscillators and Fock vacuum

The auxiliary Majorana fermion has modes f_r ; this is the fermion denoted by ψ_r in the human note. The free-field realization of the SCA uses a Heisenberg oscillator c_m^ϵ , an NS fermion ϑ_r^ϵ , and a bosonic zero mode $\mathcal{P}$. Here $\epsilon \in \{+1, -1\}$ labels two *alternative reflected realizations* of the same SCA module; the $+$ and $-$ oscillators are not two simultaneous free fields. Their nonzero brackets are

$$\begin{aligned} [c_m^\epsilon, c_n^\epsilon] &= m\delta_{m+n,0}, & \{\vartheta_r^\epsilon, \vartheta_s^\epsilon\} &= \delta_{r+s,0}, & \{f_r, f_s\} &= \delta_{r+s,0}, \\ [\mathcal{P}, c_m^\epsilon] &= [\mathcal{P}, \vartheta_r^\epsilon] = 0, & [f_r, c_m^\epsilon] &= 0, & \{f_r, \vartheta_s^\epsilon\} &= 0. \end{aligned} \quad (2)$$

The auxiliary BPZ convention inherited from the human note is

$$f_r^\dagger = -f_{-r}. \quad (3)$$

Let $\mathbf{1}_F$ be annihilated by f_r for $r > 0$, and let $|P\rangle_\epsilon$ be annihilated by c_m^ϵ and ϑ_r^ϵ for $m, r > 0$, with $\mathcal{P}|P\rangle_\epsilon = P|P\rangle_\epsilon$. We write

$$|\text{vac}; P\rangle := \mathbf{1}_F \otimes |P\rangle_\epsilon. \quad (4)$$

Thus the tensor-product module is the span of arbitrary finite products of the negative modes f_{-r} , c_{-m}^ϵ , and ϑ_{-r}^ϵ acting on $|\text{vac}; P\rangle$.

1.2 SCA generators

In the realization labelled by ϵ , the abstract SCA generators act for $m \neq 0$ as

$$\begin{aligned} L_m &= \frac{1}{2} \sum_{j \neq 0, m} c_j^\epsilon c_{m-j}^\epsilon + \frac{1}{2} \sum_{r \in \mathbb{Z} + \frac{1}{2}} \left(r - \frac{m}{2} \right) \vartheta_{m-r}^\epsilon \vartheta_r^\epsilon + \frac{i}{2} (Qm + 2\epsilon\mathcal{P}) c_m^\epsilon, \\ G_r &= \sum_{m \neq 0} c_m^\epsilon \vartheta_{r-m}^\epsilon + i(Qr + \epsilon\mathcal{P}) \vartheta_r^\epsilon, \\ L_0 &= \sum_{m > 0} c_{-m}^\epsilon c_m^\epsilon + \sum_{r > 0} r \vartheta_{-r}^\epsilon \vartheta_r^\epsilon + \frac{1}{2} \left(\frac{Q^2}{4} - \mathcal{P}^2 \right). \end{aligned} \quad (5)$$

These formulas obey the $\mathcal{N} = 1$ SCA relations at the central charge in Eq. (1), and $L_0|P\rangle_\epsilon = h(P)|P\rangle_\epsilon$. The realization $\epsilon = -1$ is used for $k > 0$, and $\epsilon = +1$ for $k < 0$. They are related by the super-Liouville reflection operator.

It is now meaningful to define the isotropic fermionic modes

$$\chi_r^\epsilon := f_r - i\vartheta_r^\epsilon, \quad \{\chi_r^\epsilon, \chi_s^\epsilon\} = 0. \quad (6)$$

The vanishing anticommutator is the reason that the highest vector below is a finite Fermi-sea product.

1.3 The commuting Virasoro pair

For completeness, define the auxiliary-fermion stress tensor and the mixed bilinear

$$L_n^F = \frac{1}{2} \sum_{r \in \mathbb{Z} + \frac{1}{2}} r : f_{n-r} f_r :, \quad U_n = \sum_{r \in \mathbb{Z} + \frac{1}{2}} f_{n-r} G_r. \quad (7)$$

The two commuting Virasoro generators used in the human note are

$$\begin{aligned} L_n^{(1)} &= \frac{b^{-1}}{b^{-1} - b} L_n - \frac{b^{-1} + 2b}{b^{-1} - b} L_n^F + \frac{1}{b^{-1} - b} U_n, \\ L_n^{(2)} &= \frac{b}{b - b^{-1}} L_n - \frac{b + 2b^{-1}}{b - b^{-1}} L_n^F + \frac{1}{b - b^{-1}} U_n. \end{aligned} \quad (8)$$

They obey

$$[L_m^{(1)}, L_n^{(2)}] = 0, \quad c^{(i)} = 1 + 6(Q^{(i)})^2, \quad Q^{(i)} = b^{(i)} + (b^{(i)})^{-1}, \quad (9)$$

where

$$b^{(1)} = \frac{2b}{\sqrt{2 - 2b^2}}, \quad (b^{(2)})^{-1} = \frac{2b^{-1}}{\sqrt{2 - 2b^{-2}}}. \quad (10)$$

This finishes the algebraic setup. Every branching vector computed below is an explicit vector in the Fock space of Eqs. (2)–(4).

2 The quantity left open in the human note

The human note uses

$$\mathcal{M}_{\text{NS}}(h) = \mathcal{V}_{F_{1/2}} \otimes \mathcal{V}_{\text{SCA}_{1/2}}(h), \quad h = \frac{1}{2} \left(\frac{Q^2}{4} - P^2 \right), \quad Q = b + b^{-1}, \quad (11)$$

with the auxiliary free-fermion factor first. The branching rule is

$$\mathcal{M}_{\text{NS}}(h) \cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} \mathcal{V}_{c^{(1)}}(h_n^{(1)}) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)}), \quad h_n^{(1)} + h_n^{(2)} = h + 2n^2. \quad (12)$$

The corresponding highest-weight vector is denoted by v_n , and its parity is $2n \pmod{2}$.

For an ordered trinion, the human note defines

$$B_a(h_3, h_2, h_1; n_3, n_2, n_1) = \frac{\widehat{\rho}_a(v_{3,n_3}, v_{2,n_2}, v_{1,n_1})}{\|v_{3,n_3}\| \|v_{2,n_2}\| \|v_{1,n_1}\|}, \quad a = 2(n_1 + n_2 + n_3) \pmod{2}. \quad (13)$$

Only B_a^2 enters the conformal-block decomposition. Our target is therefore

$$\mathcal{B}_a := B_a^2 = \frac{T_a^2}{N_{k_3}(P_3)N_{k_2}(P_2)N_{k_1}(P_1)}, \quad k_i := 2n_i, \quad (14)$$

where

$$T_a := \hat{\rho}_a(v_{3,n_3}, v_{2,n_2}, v_{1,n_1}), \quad N_k(P) := \langle P, k \mid P, k \rangle. \quad (15)$$

The order in Eq. (13) is bra, inserted field, ket. Other slot orders are obtained with the Mobius and Koszul conventions already fixed in the human note.

3 Highest vector from the free-field realization

3.1 The Fermi-sea vector

Set $k = 2n \in \mathbb{Z}$. For $k \neq 0$, the prescription is

$$|P, k\rangle = \Omega_k(P) \prod_{j=1}^{|k|} \chi_{-(2j-1)/2}^{-\text{sgn}(k)} |\text{vac}; P\rangle, \quad |P, 0\rangle = |\text{vac}; P\rangle. \quad (16)$$

The product is ordered with increasing j from left to right. Its total level is

$$\sum_{j=1}^{|k|} \frac{2j-1}{2} = \frac{k^2}{2} = 2n^2, \quad (17)$$

which is precisely the shift in Eq. (12).

The factor $\Omega_k(P)$ is fixed by the standard SCA-PBW normalization

$$|P, k\rangle = \left(G_{-1/2}^{k^2} + \text{terms of lower degree in } G \right) |P\rangle. \quad (18)$$

This fixes the vector, including its scale, without referring to a Whittaker state.

The commutators quoted in the literature are

$$\begin{aligned} [L_m^{(1)} + L_m^{(2)}, \chi_r^\epsilon] &= -\left(\frac{m}{2} + r\right) \chi_{m+r}^\epsilon, \\ [bL_m^{(1)} + b^{-1}L_m^{(2)}, \chi_r^\epsilon] &= -((m+r)Q + \epsilon\mathcal{P}) \chi_{m+r}^\epsilon + i \sum_{j \neq 0} c_j^\epsilon \chi_{m+r-j}^\epsilon. \end{aligned} \quad (19)$$

They prove directly that Eq. (16) is killed by both positive Virasoro algebras. The two weights are

$$\begin{aligned} h_n^{(1)} &= \frac{1}{4}(Q^{(1)})^2 - \left(\frac{P}{\sqrt{2-2b^2}} + nb^{(1)}\right)^2, \\ h_n^{(2)} &= \frac{1}{4}(Q^{(2)})^2 - \left(\frac{P}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1}\right)^2. \end{aligned} \quad (20)$$

The first nontrivial vectors, already converted back to the human note's SCA and auxiliary-fermion modes, are

$$\begin{aligned} |P, 1\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} + P\right) f_{-1/2} \right] |P, 0\rangle, \\ |P, -1\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} - P\right) f_{-1/2} \right] |P, 0\rangle. \end{aligned} \quad (21)$$

They are the human-note vectors $v_{1/2}$ and $v_{-1/2}$.

4 Factorized three-point numerator

4.1 Dictionary for an ordered trinion

Parameterize the inserted NS weight by

$$h_2 = \frac{1}{2}\alpha_2(Q - \alpha_2), \quad \alpha_2 = \frac{Q}{2} + P_2. \quad (22)$$

For the ordered matrix element

$$\langle P_3, k_3 | \mathbb{V}_{\alpha_2}^{(k_2)}(1) | P_1, k_1 \rangle, \quad (23)$$

the state-field correspondence identifies $\mathbb{V}_{\alpha_2}^{(k_2)}$ with the same highest vector $|P_2, k_2\rangle$ used on the middle leg. The normalized ratio in arXiv:1111.2803 is

$$l(\alpha, m | P', k', P, k) = \begin{cases} \frac{\langle P', k' | \mathbb{V}_{\alpha}^{(m)} | P, k \rangle}{\langle P' | \mathbb{V}_{\alpha}^{(0)} | P \rangle}, & m + k + k' \text{ even}, \\ \frac{\langle P', k' | \mathbb{V}_{\alpha}^{(m)} | P, k \rangle}{\langle P' | \mathbb{V}_{\alpha}^{(\pm 1)} | P \rangle}, & m + k + k' \text{ odd}. \end{cases} \quad (24)$$

The denominators are exactly the human note's kinematic normalizations: $\rho_0(\phi_3, \phi_2, \phi_1) = 1$ in the even case and $\rho_1(\phi_3, G_{-1/2}\phi_2, \phi_1) = 1$ in the odd case. Therefore

$$\boxed{T_a = l(\alpha_2, k_2 | P_3, k_3, P_1, k_1), \quad a = k_1 + k_2 + k_3 \pmod{2}.} \quad (25)$$

4.2 The product formula

For $\sigma, \tau \in \{+1, -1\}$, define

$$x_{\sigma\tau} := \alpha + \sigma P' + \tau P, \quad r_{\sigma\tau} := \frac{m + \sigma k' + \tau k}{2}. \quad (26)$$

For a nonnegative integer r , set

$$\begin{aligned} s_{\text{even}}(x, r) &= 2^{-r^2/2} \prod_{\substack{i, j \geq 1, \ i+j \leq 2r \\ i+j \equiv 0 \pmod{2}}} [x + (i-1)b + (j-1)b^{-1}], \\ s_{\text{odd}}(x, r) &= 2^{-r(r+1)/2} \prod_{\substack{i, j \geq 1, \ i+j \leq 2r+1 \\ i+j \equiv 1 \pmod{2}}} [x + (i-1)b + (j-1)b^{-1}]. \end{aligned} \quad (27)$$

For negative integer r , extend them by

$$s_{\text{even}}(x, r) = (-1)^r s_{\text{even}}(Q - x, -r), \quad s_{\text{odd}}(x, r) = s_{\text{odd}}(Q - x, -r). \quad (28)$$

Finally define

$$\text{Int}(x) := \text{sgn}(x) \lfloor |x| \rfloor. \quad (29)$$

The numerator is then

$$\boxed{l(\alpha, m | P', k', P, k) = \begin{cases} \prod_{\sigma, \tau = \pm 1} s_{\text{even}}(x_{\sigma\tau}, r_{\sigma\tau}), & m + k + k' \text{ even}, \\ \prod_{\sigma, \tau = \pm 1} s_{\text{odd}}(x_{\sigma\tau}, \text{Int}(r_{\sigma\tau})), & m + k + k' \text{ odd}. \end{cases}} \quad (30)$$

This is already the desired unnormalized three-point coefficient of the three highest vectors.

5 Norm from the same formula

Take the inserted field to be the identity:

$$\alpha = 0, \quad m = 0, \quad P' = P, \quad k' = k. \quad (31)$$

Then Eq. (24) becomes the norm, while the four factors in Eq. (30) reduce to

$$(x_{++}, r_{++}) = (2P, k), \quad (x_{+-}, r_{+-}) = (0, 0), \quad (x_{-+}, r_{-+}) = (0, 0), \quad (x_{--}, r_{--}) = (-2P, -k). \quad (32)$$

Since $s_{\text{even}}(x, 0) = 1$, we obtain

$$\boxed{N_k(P) = \langle P, k | P, k \rangle = s_{\text{even}}(2P, k) s_{\text{even}}(-2P, -k).} \quad (33)$$

Equivalently, for $p \geq 0$,

$$\begin{aligned} N_p(P) &= (-1)^p s_{\text{even}}(2P, p) s_{\text{even}}(Q + 2P, p), \\ N_{-p}(P) &= (-1)^p s_{\text{even}}(Q - 2P, p) s_{\text{even}}(-2P, p). \end{aligned} \quad (34)$$

At level $1/2$, Eq. (21) gives an independent direct check. Using

$$\langle G_{-1/2}\phi, G_{-1/2}\phi \rangle = 2h, \quad \langle f_{-1/2}\mathbf{1}, f_{-1/2}\mathbf{1} \rangle = -1, \quad (35)$$

and vanishing mixed inner product, one finds

$$\begin{aligned} N_1(P) &= 2h - \left(\frac{Q}{2} + P\right)^2 = -P(Q + 2P), \\ N_{-1}(P) &= 2h - \left(\frac{Q}{2} - P\right)^2 = P(Q - 2P). \end{aligned} \quad (36)$$

These agree with Eq. (33), because $s_{\text{even}}(x, 1) = x/\sqrt{2}$. The same computation also gives

$$\langle P, 1 | P, -1 \rangle = 0. \quad (37)$$

6 Closed formula for the human-note coefficient

Combining Eqs. (14), (25), and (33) gives

$$\boxed{\mathcal{B}_a(P_3, P_2, P_1; k_3, k_2, k_1) = \frac{l \left(\frac{Q}{2} + P_2, k_2 \mid P_3, k_3, P_1, k_1\right)^2}{N_{k_3}(P_3) N_{k_2}(P_2) N_{k_1}(P_1)},} \quad (38)$$

where l is Eq. (30), N is Eq. (33), and

$$k_i = 2n_i, \quad a = k_1 + k_2 + k_3 \pmod{2}. \quad (39)$$

This is the coefficient that multiplies the product of the two Virasoro blocks in the all-NS decomposition of the human note.

If an unsquared B_a is desired, one must choose square roots of the three complex Shapovalov norms and fix an overall sign for every branch vector. The sewn conformal block is insensitive to that choice and naturally uses Eq. (38).

7 Low-level values

The product formula immediately gives

$$l(\alpha, 0 \mid P', 0, P, 0) = 1, \quad l(\alpha, 1 \mid P', 0, P, 0) = 1. \quad (40)$$

The first is the even primary normalization; the second is the odd normalization.

With one $k = 1$ branch vector and the other two labels zero,

$$l(\alpha, 0 \mid P', 0, P, 1) = l(\alpha, 0 \mid P', 1, P, 0) = 1. \quad (41)$$

Consequently the corresponding squared normalized coefficient is simply the inverse norm of that leg.

With two $k = 1$ vectors, the even numerators are

$$\begin{aligned} l(\alpha, 0 \mid P', 1, P, 1) &= \frac{1}{2}(\alpha + P' + P)(\alpha - P' - P - Q), \\ l(\alpha, 1 \mid P', 1, P, 0) &= \frac{1}{2}(\alpha + P' + P)(\alpha + P' - P), \\ l(\alpha, 1 \mid P', 0, P, 1) &= \frac{1}{2}(\alpha + P' + P)(\alpha - P' + P). \end{aligned} \quad (42)$$

These are the first nontrivial polynomial checks to reproduce directly from the human-note Ward identities.

For example,

$$\mathcal{B}_0(P_3, P_2, P_1; 1, 0, 1) = \frac{(\alpha_2 + P_3 + P_1)^2(\alpha_2 - P_3 - P_1 - Q)^2}{4N_1(P_3)N_1(P_1)}, \quad \alpha_2 = \frac{Q}{2} + P_2. \quad (43)$$

7.1 Complete unpacking for $n_i = 0, \pm\frac{1}{2}$

We now remove the product notation completely at the first nontrivial branching level. In this subsection the slot order (n_1, n_2, n_3) is ket, inserted field, bra. To keep the left-hand sides readable, define only

$$\mathcal{B}_a[n_1, n_2, n_3] := \mathcal{B}_a(P_3, P_2, P_1; 2n_3, 2n_2, 2n_1), \quad T_a[n_1, n_2, n_3] := \hat{\rho}_a(v_{3,n_3}, v_{2,n_2}, v_{1,n_1}). \quad (44)$$

No momentum argument is suppressed on the right-hand sides below.

Whenever $n_i \neq 0$, write

$$\epsilon_i := 2n_i \in \{+1, -1\}. \quad (45)$$

The free-field vector and its norm take the uniform form

$$\begin{aligned} |P, \epsilon\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} + \epsilon P \right) f_{-1/2} \right] |P, 0\rangle, \\ \langle P, \epsilon | P, \epsilon \rangle &= -\epsilon P (Q + 2\epsilon P). \end{aligned} \quad (46)$$

Zero or one nonzero label. For the vacuum branch,

$$T_0[0, 0, 0] = 1, \quad \mathcal{B}_0[0, 0, 0] = 1. \quad (47)$$

If exactly one label is nonzero, the numerator remains one and

$$\begin{aligned} T_1[\epsilon_1/2, 0, 0] &= T_1[0, \epsilon_2/2, 0] = T_1[0, 0, \epsilon_3/2] = 1, \\ \mathcal{B}_1[\epsilon_1/2, 0, 0] &= -\frac{\epsilon_1}{P_1(Q + 2\epsilon_1 P_1)}, \\ \mathcal{B}_1[0, \epsilon_2/2, 0] &= -\frac{\epsilon_2}{P_2(Q + 2\epsilon_2 P_2)}, \\ \mathcal{B}_1[0, 0, \epsilon_3/2] &= -\frac{\epsilon_3}{P_3(Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (48)$$

For example, a $+\frac{1}{2}$ label gives $-[P_i(Q + 2P_i)]^{-1}$, whereas a $-\frac{1}{2}$ label gives $[P_i(Q - 2P_i)]^{-1}$.

Exactly two nonzero labels. When the ket label vanishes, one finds

$$\begin{aligned} T_0[0, \epsilon_2/2, \epsilon_3/2] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right], \\ \mathcal{B}_0[0, \epsilon_2/2, \epsilon_3/2] &= \frac{\left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right]^2}{4\epsilon_2 \epsilon_3 P_2 P_3 (Q + 2\epsilon_2 P_2) \times (Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (49)$$

When the inserted label vanishes, the ordered trinion gives instead

$$\begin{aligned} T_0[\epsilon_1/2, 0, \epsilon_3/2] &= \frac{1}{2} \left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3 \right)^2 \right], \\ \mathcal{B}_0[\epsilon_1/2, 0, \epsilon_3/2] &= \frac{\left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3 \right)^2 \right]^2}{4\epsilon_1 \epsilon_3 P_1 P_3 (Q + 2\epsilon_1 P_1) \times (Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (50)$$

Finally, when the bra label vanishes,

$$\begin{aligned} T_0[\epsilon_1/2, \epsilon_2/2, 0] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right], \\ \mathcal{B}_0[\epsilon_1/2, \epsilon_2/2, 0] &= \frac{\left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right]^2}{4\epsilon_1 \epsilon_2 P_1 P_2 (Q + 2\epsilon_1 P_1) \times (Q + 2\epsilon_2 P_2)}. \end{aligned} \quad (51)$$

The relative sign in Eq. (50) is part of the ordered three-point prescription; it should not be replaced by a symmetric guess. It disappears after squaring the numerator but remains relevant if an unsquared coefficient is assigned.

Three nonzero labels. All eight sign choices are covered by

$$T_1[\epsilon_1/2, \epsilon_2/2, \epsilon_3/2] = \frac{1}{2} \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b \right) \times \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b^{-1} \right), \quad (52)$$

$$\mathcal{B}_1[\epsilon_1/2, \epsilon_2/2, \epsilon_3/2] = - \frac{\epsilon_1 \epsilon_2 \epsilon_3 \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b \right)^2 \times \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b^{-1} \right)^2}{4 P_1 P_2 P_3 (Q + 2\epsilon_1 P_1)(Q + 2\epsilon_2 P_2) \times (Q + 2\epsilon_3 P_3)}. \quad (53)$$

Equations (47)–(53) exhaust all 27 choices $n_i \in \{0, \pm \frac{1}{2}\}$. They are precisely the formulas evaluated by `Code/two_virasoro_fusion.py`.

8 What remains to be audited

The calculation above settles the all-NS coefficient, modulo a direct audit of the literature product formula against the human note’s ordered Ward identities. The efficient audit is:

1. reproduce Eqs. (36) from the Gram form;
2. reproduce Eqs. (41) and Eq. (42) from the three-point Ward identities;
3. check a $|k| = 2$ case using the explicit level-two vectors in Appendix B of arXiv:1111.2803; and
4. only then use Eq. (38) as the all-level input to the block decomposition.

The Ramond branching vectors have an analogous free-field prescription in arXiv:1608.02568. It uses integer $\chi_r^\pm$ modes and has two parity copies. Their norms and (NS, R, R) three-point coefficients require a separate translation to the human note’s phase convention

$$G_0 w_\beta^\pm = i\beta e^{\mp i\pi/4} w_\beta^\mp. \quad (54)$$

No Whittaker coefficient is used here.

9 Sources

The free-field vector, its PBW normalization, the blow-up factors, and the functions $s_{\text{even}}, s_{\text{odd}}$ are in arXiv:1111.2803, especially Sec. 3.2, Proposition 3.2, and Appendix B. The later presentation and the Ramond analogue are in arXiv:1608.02568, Sec. 4.